S5 Table.

Prior work and how this study relates to it

Supporting information for "Environmental fluctuation strengthens gene regulation only when it can be sensed"

The table sets out which parts of the argument are established in the literature, which conflict with our results, and which we did not find precedent for. Literature searches covered theory of plasticity and cue reliability, gene-regulatory-network evolution under fluctuating selection, digital evolution (Avida, Aevol), network sparseness and modularity, and weighted network measures.

Claim	Status	Sources and relation to this study
Reliable cues favour plastic responses; without them, fluctuation favours generalists, bet-hedging or genetic tracking	Established	Levins 1968 [1]; Moran 1992 [2]; Kussell and Leibler 2005 [3]; Botero et al. 2015 [4]. Our results concern the corresponding effect on the regulatory network.
In slowly changing environments populations track the environment genetically instead of evolving plasticity	Established	Botero et al. 2015 [4]; Rago et al. 2019 [21]. Consistent with the better match to the target, without a change in regulation, at coarse grain in R-Aevol.
In a gene regulatory network model, plasticity evolves mainly under fast, erratic change with reliable cues	Established	Jorritsma and van den Berg 2026 [5]. Their abstract reports which parts of the network become sensitive to mutation. We have not been able to read the full text; if it reports connection strength by cue reliability, part of our architectural result would overlap with it.
In a network with a dedicated sensor gene, discrete plasticity evolves as the sensor comes to activate some genes and repress others	Published	Sakamoto and Innan 2024 [19]. The sensor is always informative and gene number does not evolve, so neither an uninformative-cue control nor genome size is examined. EvoSim's sensor node follows the same design.
Fluctuating environments can select for or against gene regulation	Established experimentally	Pollack, Nozoe and Kussell 2025 [6], who report that catabolite symport flux strongly affects selection for regulation and note that laboratory evolution under metabolic fluctuation often loses the regulation of metabolic genes.
Fluctuation without information about the optimum promotes complex regulatory networks	Published; conflicts with our R-Aevol results for a random-walk optimum	Tsuda and Kawata 2010 [7]. Their optimum moved across generations, and the networks received an on/off signal that carried no information about its position. The effect was strong for a random walk and slight, limited to large and infrequent shifts, for an optimum alternating between two positions, which is the pattern of our design. Our grain experiment found that grain does not reconcile the two results. The remaining candidate explanations are the novel optima of the random walk and differences in the genotype–phenotype map.
Fluctuating environments favour larger genomes	Published; matches earlier EvoSim results	Bentkowski et al. 2015 [8]. R-Aevol shows the opposite.

Claim	Status	Sources and relation to this study
Genome size is limited by selection for mutational robustness and falls with population size and mutation rate	Established in Aevol	Knibbe et al. 2007 [9]; Luiselli et al. 2024 [10]. Consistent with the small genomes in large EvoSim populations.
Sensing and non-sensing populations evolve differently in fluctuating environments	Published, different measures	Lalejini et al. 2021 [11] (Avida), measuring trait loss and the accumulation of deleterious mutations. Network architecture was not measured.
Network models with and without a cue input compared	Published	Ng and Kinjo 2024 [20]; plasticity-led evolution appears only with the cue input, and network density is fixed by the mutation operator. Removing the cue changes its presence and its information together, the confound the yoked control removes.
Regulatory complexity is measured by counting genes or edges	Common practice	Tsuda and Kawata 2010 [7] (gene numbers, degree distributions); Espinosa-Soto 2018 [12] (binary edges, weights not discussed).
Network size and connectivity can grow without selection for them	Established	Soyer and Bonhoeffer 2006 [13].
The number of transcription factors scales faster than genome size and is higher in free-living bacteria	Established; two interpretations	Comparative genomics links it to lifestyle and environmental variety (van Nimwegen 2003 [16]; Cases, de Lorenzo and Ouzounis 2003 [17]). R-Aevol simulations reproduce the scaling and attribute it to mutation and rearrangement rates instead of lifestyle (Beslon et al. 2010 [18]).
Summed edge strength as a network measure	Standard method	Iwasaki, Tsuda and Kawata 2013 [14].
The yoked control	Established in the study of learning	Church 1964 [15] describes a bias that arises when individual subjects are paired; our arms are not paired, so it does not apply.
Not found in the literature		
Edge count does not track edge strength under fluctuation	This study	EvoSim (exploratory; more edges with the same total strength), R-Aevol informative-cue populations (pre-registered secondary analysis; no more links but stronger regulation).
A yoked control that separates a cue's information from its presence in the evolution of regulatory networks	This study	R-Aevol, with the rates of target and signal change matched between arms (S2 Text). Varying cue reliability achieves the same separation if it includes a cue that is present but uninformative; the abstract of [5] does not say whether it does.
Regulatory strength and genome size compared across informative, yoked and absent cues	This study	Pre-registered predictions P1–P4; subject to the caveat on [5] above.

References

1. Levins R. Evolution in Changing Environments. Princeton: Princeton University Press; 1968.
2. Moran NA. The evolutionary maintenance of alternative phenotypes. Am Nat. 1992;139(5):971–89.
3. Kussell E, Leibler S. Phenotypic diversity, population growth, and information in fluctuating environments. Science. 2005;309(5743):2075–8.
4. Botero CA, Weissing FJ, Wright J, Rubenstein DR. Evolutionary tipping points in the capacity to adapt to

environmental change. Proc Natl Acad Sci USA. 2015;112(1):184–9.

5. Jorritsma A, van den Berg P. The evolution of plasticity and evolvability in a simple gene regulatory network. Evolution. 2026;80(1):230–9.
6. Pollack D, Nozoe T, Kussell E. Gene regulation, phenotypic memory, and selection in fluctuating environments. bioRxiv [Preprint]. 2025. doi:10.1101/2025.06.23.661119.
7. Tsuda ME, Kawata M. Evolution of gene regulatory networks by fluctuating selection and intrinsic constraints. PLoS Comput Biol. 2010;6(8):e1000873.
8. Bentkowski P, van Oosterhout C, Mock T. A model of genome size evolution for prokaryotes in stable and fluctuating environments. Genome Biol Evol. 2015;7(8):2344–51.
9. Knibbe C, Coulon A, Mazet O, Fayard JM, Beslon G. A long-term evolutionary pressure on the amount of noncoding DNA. Mol Biol Evol. 2007;24(10):2344–53.
10. Luiselli J, Rouzaud-Cornabas J, Lartillot N, Beslon G. Genome streamlining: effect of mutation rate and population size on genome size reduction. Genome Biol Evol. 2024;16(12):evae250.
11. Lalejini A, Ferguson AJ, Grant NA, Ofria C. Adaptive phenotypic plasticity stabilizes evolution in fluctuating environments. Front Ecol Evol. 2021;9:715381.
12. Espinosa-Soto C. On the role of sparseness in the evolution of modularity in gene regulatory networks. PLoS Comput Biol. 2018;14(5):e1006172.
13. Soyer OS, Bonhoeffer S. Evolution of complexity in signaling pathways. Proc Natl Acad Sci USA. 2006;103(44):16337–42.
14. Iwasaki WM, Tsuda ME, Kawata M. Genetic and environmental factors affecting cryptic variations in gene regulatory networks. BMC Evol Biol. 2013;13:91.
15. Church RM. Systematic effect of random error in the yoked control design. Psychol Bull. 1964;62:122–31.
16. van Nimwegen E. Scaling laws in the functional content of genomes. Trends Genet. 2003;19(9):479–84.
17. Cases I, de Lorenzo V, Ouzounis CA. Transcription regulation and environmental adaptation in bacteria. Trends Microbiol. 2003;11(6):248–53.
18. Beslon G, Parsons DP, Sanchez-Dehesa Y, Peña JM, Knibbe C. Scaling laws in bacterial genomes: a side-effect of selection of mutational robustness? BioSystems. 2010;102(1):32–40.
19. Sakamoto T, Innan H. On the evolutionary origin of discrete phenotypic plasticity. G3 (Bethesda). 2024;14(9):jkae144.
20. Ng ETH, Kinjo AR. Plasticity-led and mutation-led evolutions are different modes of the same developmental gene regulatory network. PeerJ. 2024;12:e17102.
21. Rago A, Kouvaris K, Uller T, Watson R. How adaptive plasticity evolves when selected against. PLoS Comput Biol. 2019;15(3):e1006260.